Packet Tracer - Configure Multiarea OSPFv3

Addressing Table

Device	Interface	IPv6 Address	OSPF Area
RA	G0/0/0	2001:db8:1:a1::1/64	1
	G0/0/1	2001:db8:1:a2::1/64	1
	S0/1/0	2001:db8:1:ab::2/64	0
	Link-Local	fe80::a	N/A
RB	G0/0/0	2001:db8:1:b1::1/64	0
	S0/1/0	2001:db8:1:ab::1/64	0
	S0/1/1	2001:db8:1:bc::1/64	0
	Link-Local	fe80::b	N/A
RC	G0/0/0	2001:db8:1:c1::1/64	2
	G0/0/1	2001:db8:1:c2::1/64	2
	S0/1/1	2001:db8:1:bc::2/64	0
	Link-Local	fe80::c	N/A

Objectives

Part 1: Configure OSPFv3

Part 2: Verify Multiarea OSPFv3 Operations

Background / Scenario

In this activity, you will configure multiarea OSPFv3. The network is already connected and interfaces are configured with IPv6 addressing. Your job is to enable multiarea OSPFv3, verify connectivity and examine the operation of multiarea OSPFv3.

Instructions

Part 1: Configure OSPFv3

Step 1: Enable IPv6 routing and configure OSPFv3 on RA.

a. Enable IPv6 routing.

b. Configure OSPFv3 on RA with a process ID of 1 and a router ID of 1.1.1.1.

Step 2: Advertise each directly connected network in OSPFv3 on RA.

Configure each active IPv6 interface with OSPFv3 assigning each to the area listed in the Addressing Table.

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Step 3: Configure OSPFv3 on RB and RC.

Repeat Steps 1 and 2 for **RB** and **RC**. The router ID for RB should be 2.2.2.2 and the router ID for RC should be 3.3.3.3.

Part 2: Verify Multiarea OSPFv3 Operations

Step 1: Verify connectivity to each of the OSPFv3 areas.

From RA, ping each of the remote devices in area 0 and area 2:

- 2001:db8:1:b1::2
- 2001:db8:1:a1::2
- 2001:db8:1:a2::2
- 2001:db8:1:c1::2
- 2001:db8:1:c2::2

Step 2: Use show commands to examine OSPFv3 operation.

Use the following commands to gather information about your OSPFv3 multiarea implementation.

show ipv6 ospf

show ipv6 route

show ipv6 ospf database

show ipv6 ospf interface

show ipv6 ospf neighbor

Note: Packet Tracer output for **show ipv6 protocols** is currently not aligned with IOS 15 output. Refer to the physical equipment labs for correct **show** command output.

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